TYPES OF REACTIONS

Acids and Bases

Acids

- Contain the element H.
- Produce H^{1+} ions in solution.
- Have a sour taste.
- Turn blue litmus red.

e.g. HCl - hydrochloric
HNO_3 - nitric
H_2SO_4 - sulphuric
CH_3COOH - acetic
H_3PO_4 - phosphoric

Bases

- Any metal oxide or hydroxide.
- Produce OH^{1-} ions in solution.
- Have a soapy feel and bitter taste.
- Turn red litmus blue.

e.g.	NaOH	CaO
	$Mg(OH)_2$	NH_4OH

Note

NH_4^{1+} behaves as if it is a metal!

Acid Reactions

When an acid reacts with another substance, it produces a ***salt***.

***Salt**:* non-metal part of an acid combined with a metal or ammonium ion.

- HCl - produces ***chloride*** salts.

  e.g. $MgCl_2$, NaCl, $CuCl_2$, NH_4Cl, etc.

- HNO_3 - produces ***nitrate*** salts.

  e.g. $Mg(NO_3)_2$, $NaNO_3$, $Cu(NO_3)_2$, NH_4NO_3, etc.

- H_2SO_4 - produces ***sulphate*** salts.

  e.g. $MgSO_4$, Na_2SO_4, $CuSO_4$, $(NH_4)_2SO_4$, etc.

- CH_3COOH - produces ***acetate*** salts.

  e.g. $Mg(CH_3COO)_2$, $NaCH_3COO$, etc.

General Equations

Acid + Metal → Salt + Hydrogen Gas

Molecular: $H_2\cancel{SO_4}{}_{(aq)} + 2\ Na_{(s)} \rightarrow Na_2\cancel{SO_4}{}_{(aq)} + H_{2\ (g)}$

Ionic: $2\ H^{1+}{}_{(aq)} + Na_{(s)} \rightarrow Na^{1+}{}_{(aq)} + H_{2\ (g)}$

Acid + Base → Salt + Water

Molecular: $H_2\cancel{SO_4}{}_{(aq)} + Na_2O_{(s)} \rightarrow Na_2\cancel{SO_4}{}_{(aq)} + H_2O_{(g)}$

Ionic: $2\ H^{1+}{}_{(aq)} + Na_2O_{(s)} \rightarrow 2\ Na^{1+}{}_{(aq)} + H_2O_{(l)}$

Acid + Carbonate / Hydrogencarbonate → Salt + Water + Carbon Dioxide

Molecular: $H_2\cancel{SO_4}{}_{(aq)} + Na_2CO_{3\ (s)} \rightarrow Na_2\cancel{SO_4}{}_{(aq)} + H_2O_{(l)} + CO_{2\ (g)}$

Ionic: $2\ H^{1+}{}_{(aq)} + Na_2CO_{3\ (s)} \rightarrow 2\ Na^{1+}{}_{(aq)} + H_2O_{(l)} + CO_{2\ (g)}$

Precipitation Reactions

When two soluble ionic solutions are mixed it is possible that one of the new combinations of ions produces an ***insoluble salt***. These types of reactions are known as ***precipitation reactions***.

To predict whether a precipitate will form, a table of ***solubility rules*** is used.

SOLUBLE
All Group I cations, NH_4^{1+}
All NO_3^{1-}, CH_3COO^{1-} (acetate), ClO_3^{1-} (chlorates)
All Cl^{1-}, Br^{1-}, I^{1-} except Ag^{1+}, Pb^{2+}, Hg^{2+}
All SO_4^{2-} except Ba^{2+}, Sr^{2+}, Pb^{2+}, Ca^{2+} slightly soluble

INSOLUBLE
All O^{2-}, OH^{1-} except Group I, NH_4^{1+}, Ba^{2+} Ca^{2+}, Sr^{2+} slightly soluble
All S^{2-} except Group I, NH_4^{1+}
All CO_3^{2-}, PO_4^{3-}, SO_3^{2-} except Group I, NH_4^{1+}

Using The Solubility Rules

Barium nitrate and sodium sulphate solutions are mixed. Will a precipitate form? If yes, write a molecular and net ionic equation to represent the reaction.

1. ***Write down the formula for each solution being mixed.*** They can both be indicated as (aq) because they are soluble:

$$Ba(NO_3)_{2\ (aq)} + Na_2SO_{4\ (aq)}$$

2. ***Look at the two possible new combinations of ions, one at a time, and check out the solubility table to see if an insoluble substance is produced.***

   The solubility rules indicate that, of the two possible products $BaSO_4$ and $NaNO_3$, the $BaSO_4$ is insoluble.

3. ***Write out an equation that identifies the insoluble substance formed with an (s).*** The other possible combination can be represented with an (aq) to indicate that it remains in solution and is soluble.

$$Ba(\cancel{NO_3)}_{2\ (aq)} + \cancel{Na_2}SO_{4\ (aq)} \rightarrow BaSO_{4\ (s)} + 2\ \cancel{NaNO_3}_{\ (aq)}$$

4. ***Write out the net ionic equation by leaving out the spectator ions on both sides of the equation.***

$$Ba^{2+}_{\ (aq)} + SO_4^{2-}{}_{(aq)} \rightarrow BaSO_{4\ (s)}$$